

FORCE AND LAWS OF MOTION

- **Force**

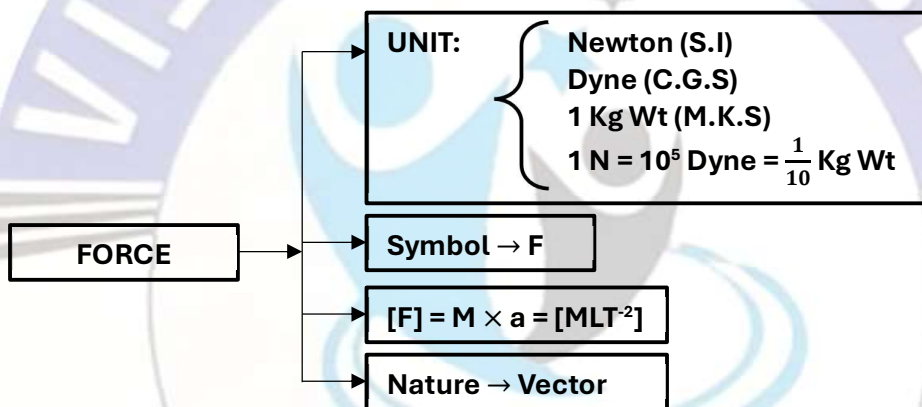
- Force is a two bodies interaction which when appears changes or, tries to change state of object or, shape of object or both.
- The factor responsible for change in velocity is force.

Special Note

1. Force can only appear when two bodies are present.
2. Force always appears in pair, one is action and another is reaction.
3. State doesn't change necessary when force applied but force try to change the state.
4. Force has direction as well as magnitude So, Force is a vector quantity.
5. Force can also be represented by arrow, and it follows all the properties of vector.
Like as Force ($\vec{F}$).

- **Properties of Force**

- Force is a physical quantity which has



- **Effect of Forces**

- There are following effects of force when it applies on a body.
 1. A Force or, a set of forces can move a stationary body.
 2. A Force or, a set of forces can stop a moving body.
 3. A Force or, a set of forces can change the speed of moving body.
 4. A Force or, a set of forces can change the direction of the moving body.
 5. A Force or, a set of forces can change the shape of the moving body.
 6. A Force or, a set of forces can change the size or, shape of a rest bodies.

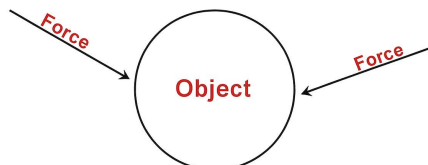
Special Note

- A Force is that physical cause changes or, try to change either the size or, shape or, the state of rest or, of motion of a body.

- **Depending on the type of arrow force has following types:**

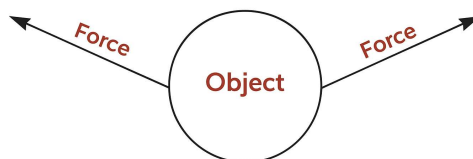
1. **Push Force**

- The head of arrow of force is on the object.



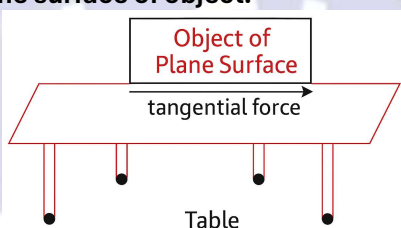
2. Pull Force

- The tail of arrow as force is on the object.



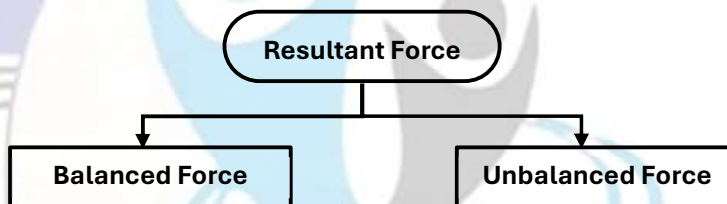
3. Tangential Force

- The arrow of force is tangential to the surface of object.



• Resultant Force

- The arithmetic sum of all forces applied on a body is called resultant force.
- It may be positive, negative or zero.



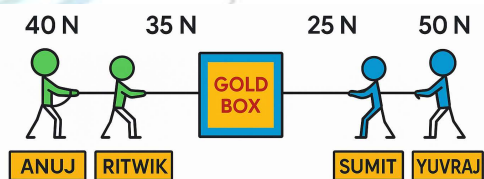
1. Balanced Force

- If the sum of all forces applied on a body is zero, then the resultant force is known as balanced force.

Left Side = $40\text{ N} + 35\text{ N} = 75\text{ N}$, Right Side = $50\text{ N} + 25\text{ N} = 75\text{ N}$

$$\sum F_{\text{net, ext}} = 75\text{ N} - 75\text{ N} = 0$$

- Thus, gold box is balanced. It is neither shift left nor right.



2. Unbalanced Force

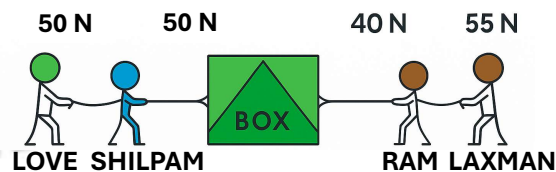
- If the sum of all forces applied on a body is non-zero, then the resultant force is known as unbalanced force.

Left side total force = $50\text{ N} + 50\text{ N} = 100\text{ N}$,

Right side total force = $40\text{ N} + 55\text{ N} = 95\text{ N}$

$$\sum F_{\text{net, ext}} = 100\text{ N} - 95\text{ N} = 5\text{ N}$$

- Thus, the diamond box is unbalanced it is shifted towards left.



Types of Forces

- From the point of view of application, Force has two kinds.

1. Contact Force.
2. Non-Contact Force.

1. Contact Force

- The Forces which are applied on bodies by making a physical contact with them, are called contact forces.

Types of Contact Forces

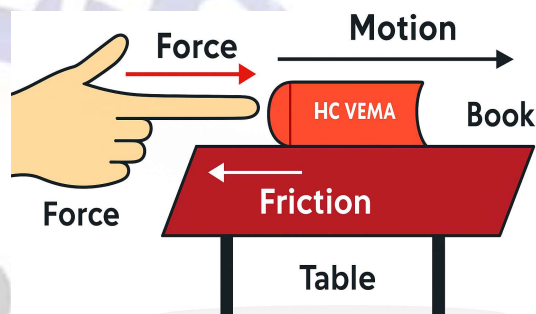
- i) Frictional Force.
- ii) Normal Force.
- iii) Tension Force.
- iv) Force exerted by spring.
- v) Force exerted during collision.

i) Frictional Force/Friction

- Friction is a force of contact that appears between surface in contact in a direction parallel to the surface and it opposes relative motion or, tendency to relative motion.

Note:

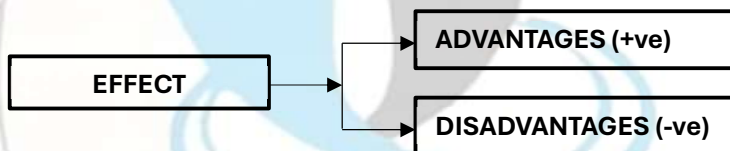
1. Friction $\propto$ roughness of surface.
2. Friction $\propto \frac{1}{\text{smoothness of surface}}$.



- Friction is also called Force of friction.

▪ Effect of Friction

- a) Oppose the slipping of the two bodies against each – other.
- b) Reduces the speed of moving objects and it even stops the motion of the object.



▪ Advantages of friction

- a) Friction help us to walk on the ground.
- b) We can write on a paper or, on a board due to friction.
- c) Friction helps in stopping a vehicle on applying the breaks.
- d) It helps in generating heat when we rub our hands together.
- e) Helpful in eating and drinking.
- f) Lighting a match stick by rubbing.
- g) It helps to climb a tree.
- h) Trains, bus, bike, cycle, etc. run on the rail/road etc.
- i) Nails and screws join two surfaces due to friction.
- j) Asteroids burn in the atmosphere before reaching the earth due to friction and saves lives on the earth.

▪ Disadvantages of Friction

- a) Friction produce a lot of heat in various parts of machinery and leads to wastage of energy as heat.
- b) Due to opposes motion, hence more energy is needed to overcome friction.
- c) Noise production in a machines is irritating as well as leads to energy loss.
- d) Forest fires are caused due to friction between branches of the tree.
- e) Due to friction, engines of auto – mobiles consume more fuel which is a loss of money.
- f) Due to friction we have to exert more power in machines.
- g) It causes wear and tear of the machines.
- h) It decreases the life expectancy of moving parts of vehicle.
- i) It decreases the efficiency of moving body.

Types of Friction

1. Static Friction.
2. Dynamic Friction.

1. Static Friction

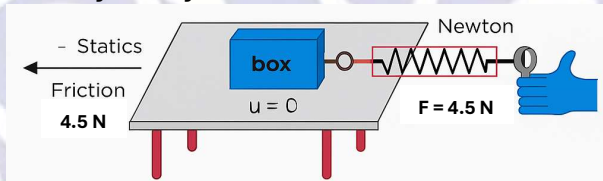
- This friction opposes tendency of relative motion between the surfaces.

Note:

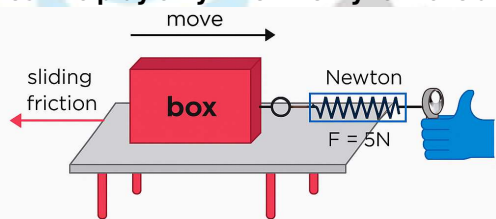
1. Static Friction appears when relative velocity between surfaces in contact is zero.

For example:-

- ✓ Statics friction acts when a person try to walk on the ground.
2. Friction experienced by a body when it is at rest is called static friction.



3. Statics friction is a kind of starting friction because an object just tends to start moving, it does not actually move.
4. Statics friction is a self adjusting force. It is just equal to pull/push force.
5. Static friction comes into play only when we try to move an stationary object.



In this case, the box is move in the direction of force because applied force (5 N) is greater than frictional forces (Maxⁿ 4.5 N).

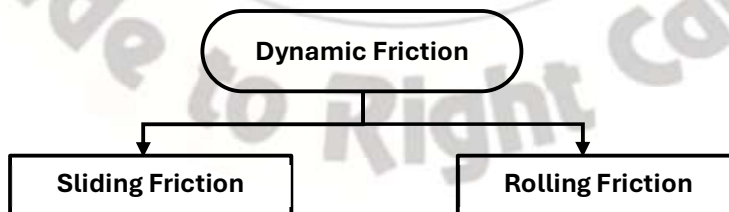
Here the frictional force created is known as sliding friction.

2. Dynamic Friction

- Friction experienced by a body when it is in motion, called dynamic friction.

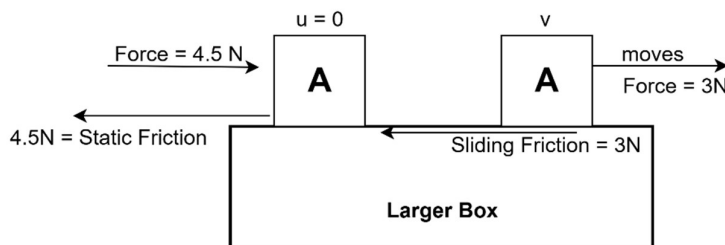
OR

This friction appears when two surfaces are contact have relative motion.



a) Sliding Friction

- The frictional force present when one object moves slowly (or slides) over the surface of another object, is known as sliding friction.
- The sliding friction comes into play when an object is sliding (moving slowly but continuously) over another object.



- Sliding friction is always less than static friction.

Explanation:

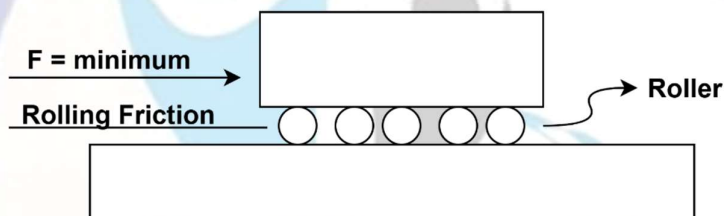
- Since the interlocking of the two surfaces is less when an object has already started moving. Therefore, the sliding friction is smaller than the static friction.

Special Note:

- Dynamic Friction is also called kinetic friction.

b) Rolling Friction

- When an object (like as wheel) rolls over the surface of another object, the resistance to its motion is called rolling friction.
- It is always easier to 'roll' than to slides an object over another object.
- So, rolling friction is much less than sliding friction.
- Thus, rolling reduces friction.



Note:

1. Statics friction $>$ sliding friction $>$ Rolling friction

For example:-

- ✓ Pull heavy object/luggage (like heavy trolley bags/suitcases etc.) easy to roll OR pulling of cars, buses etc.
- ✓ Heavy machines can be easily moved from one place to another place by rolling.

2. We conclude that when an object rolls over another objects, then rolling friction comes into play between them.
3. Wheel is a kind of rollers. It is said to be one of the greatest inventions of mankind. It is because wheels greatly reduce friction (by rolling) and hence can be used to move even heavy objects (like cars, buses, trucks and trains) rather easily.

▪ **Mechanics of Friction**

- Friction while opposing the relative motion.
 - i) Opposes the motion of object moving ahead.
 - ii) Support the motion of objects lagging behind.

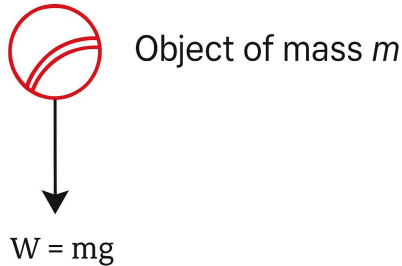
• **Weight**

- This is a force exerted by planets or Satellite on object.

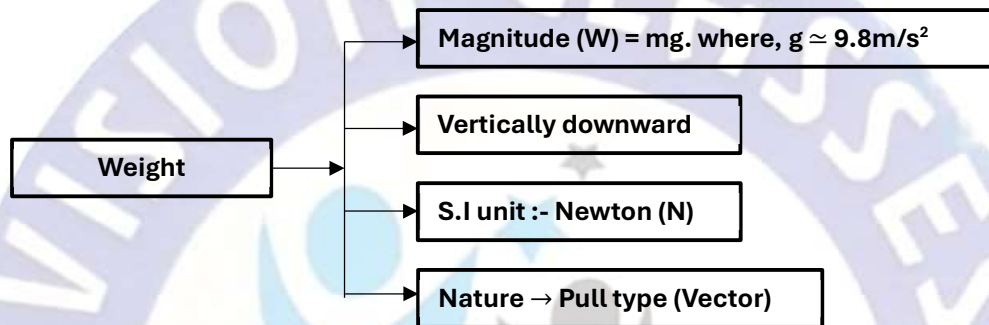
OR,

The force exerted by the earth on a body is called the weight of the body.

- This force acts towards the centre of the earth.
- If the mass of the object = M kg, Weight of the object = W



Here, Acceleration due to gravity (g) = 9.8 m/s^2
 $= 10 \text{ m/s}^2$



For example:-

Q. a person of mass 50 kg stands on the earth. Find the weight of the object.

Ans Since given a mass of a person = 50 kg
 $g = 10 \text{ m/s}^2$

$$\begin{aligned}\therefore W &= mg \\ &= 50 \times 10 \\ \Rightarrow W &= 500 \text{ N}\end{aligned}$$

Note:

1. Weight is denoted by 'W'. It is equal to Force applied on lifting object.
2. The weight of an object is not constant at every where because the value of 'g' is not constant.
3. Weight is also called force due to gravity.

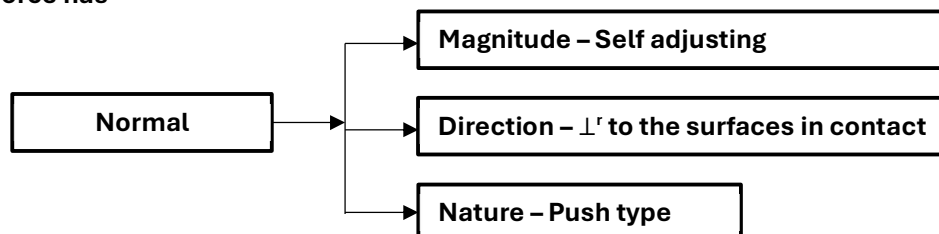
Q. If a person of mass 50 kg stands on the surface of moon. Find the weight of a person.

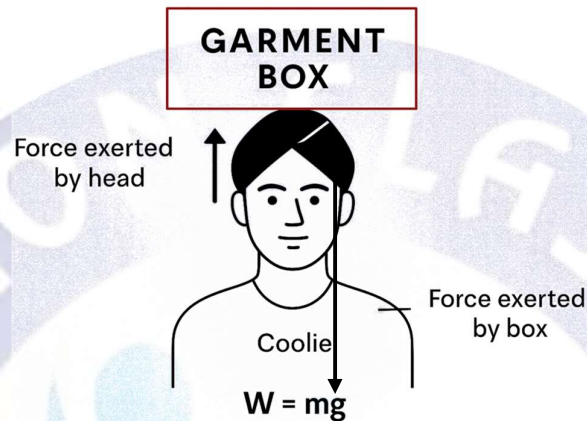
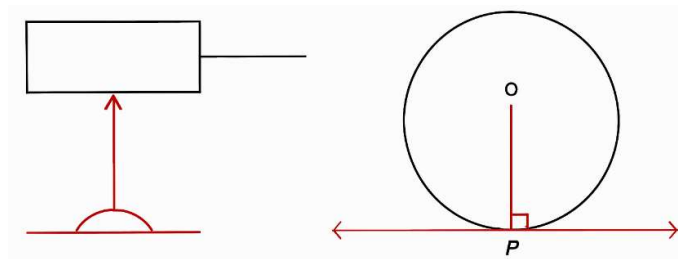
Ans ∴ The mass of person (m) = 50 kg
The value of $g = 1.63 \text{ m/s}^2$

$$\begin{aligned}\text{Weight of a person on the moon (W)} &= mg \\ &= 50 \times 1.63 \\ W &= 81.50 \text{ N}\end{aligned}$$

• **Normal OR Normal Force**

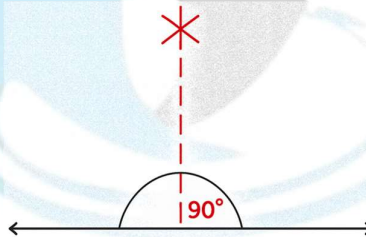
- This force appears between surfaces in contact.
- The normal force has



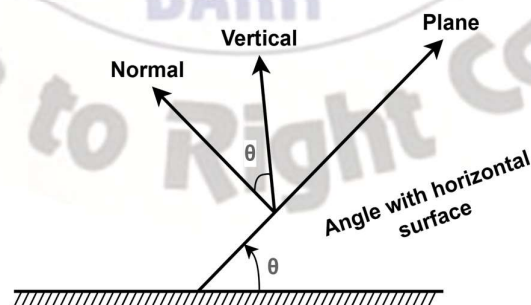


Note:

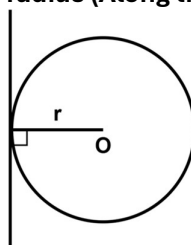
1. To draw perpendicular to make a semicircle and draw another line which divide semicircle in two equal parts.



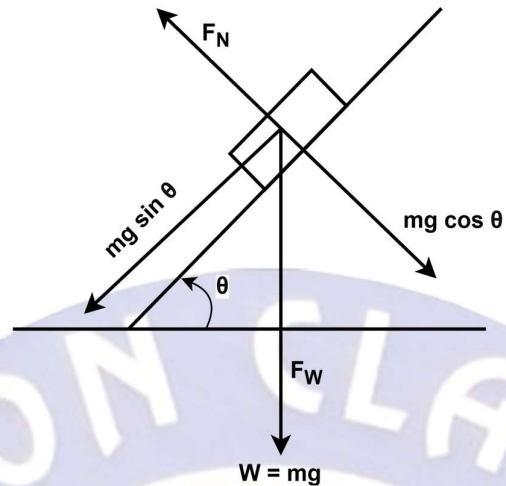
2. The direction of perpendicular can be found as.
 - a) For horizontal surface $\perp$ is vertical.
 - b) For vertical surface $\perp$ is horizontal.
 - c) If the surface make the angle θ with horizontal the $\perp$ makes angle θ with vertical and vice – versa.



- d) For spherical surface $\perp$ is radius (Along the radius).

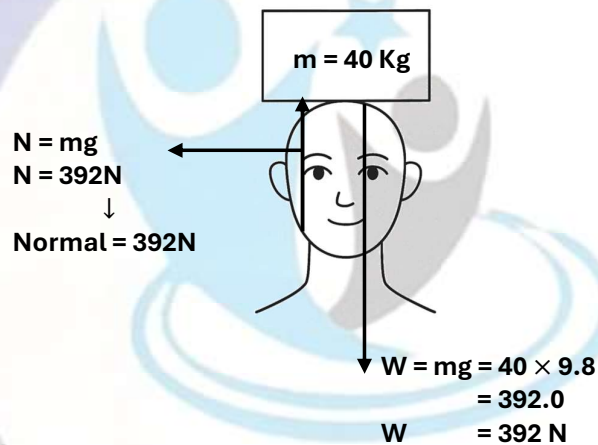


Note:



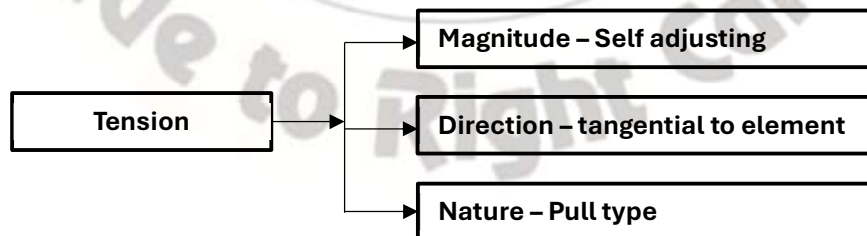
Experiment:

- If a person lifted 40 kg on his head then



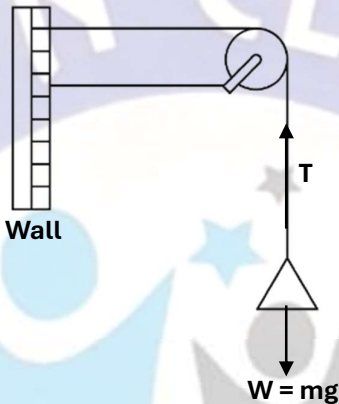
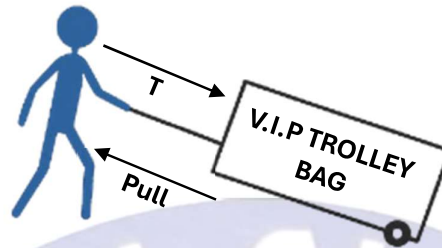
- **Tension**

- This force is exerted by linear elements like rope, string, spring, chain, rod etc.
- This force is appears when length of linear element try to increases.
- Force exerted by a string is called tension.



Note:

- Those substances which are not flexible, like rod and spring can exert a force opposite to the tension called Compression (Push type).



- The tension or, compression of spring is quantitatively (numerically) is given by Hook's law which can be states as-

$$\text{Tension or, compression} = K \times \Delta L$$

Where,

K = Force Constant
or
Spring Constant

$$0.9 \text{ N/m} - 4.8 \text{ N/m}$$

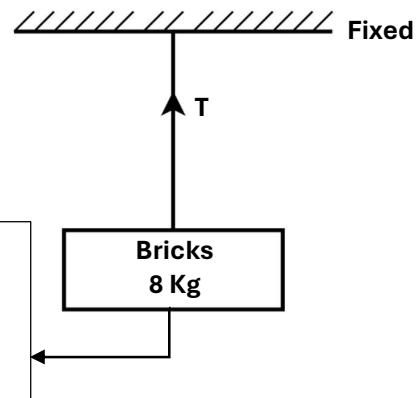
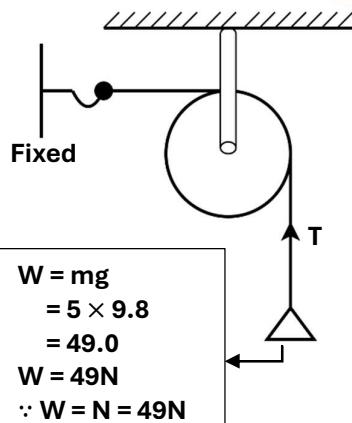
$$\Delta L = l_2 - l_1$$

'+' Sign is for tension

'-' Sign is for Compression

For example:-

Q. Find tension

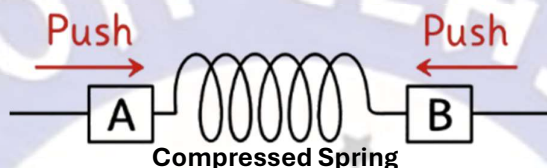


- **Force exerted by a spring**

- A spring is made up of a coiled metallic wire; the actual length of spring is called natural length of spring.
- The spring may either be pulled or pushed.
- If the spring is pulled at the end so that the length becomes larger than its natural length, it is called a stretched spring or, an extended spring.



- if the spring is pushed at the ends so that the length becomes smaller than its natural length, it is called a Compressed Spring.



Note:

- In the natural length of spring does not exert any force on the objects attached to its ends.

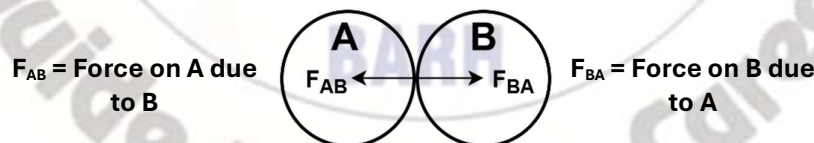
- **Force exerted during collision**

- When two bodies collide, they push each other as a result, equal and opposite forces act on each body. These forces are the Force of action and Force of reaction.

Case 1: A and B moving towards each other



Case 2: Force exerted due to collision



Case 3: A & B moving away from each other after collision



2. **Non – Contact Force**

- The forces experienced by bodies even without being physically in touch, are called non – contact forces or, Forces at a distance.

For example:-

- i) Gravitational Force.

- ii) **Electrostatic Force.**
- iii) **Magnetic Force.**

i) **Gravitational Force**

- In the universe, each particle attracts the other particle due to its mass. This force of attraction between them is called gravitational force.
- The earth also, because of its mass, attracts all other masses around it.
- The force on a body due to earth's attraction is called the force of gravity or, the weight of the body.

For example:-

- ✓ A coin falls down when it is released at a height.
- ✓ All fruits fall down from tree.

ii) **Electrostatics Force**

- Two like charges repel, while two unlike charges attract each – other, the force between the charges is called electrostatic force.
- This force acts between the charged
Like charges $\{(+, +) \text{ or } (-, -)\} \rightarrow \text{Repel}$
Unlike charges $\{(+, -) \text{ or } (-, +)\} \rightarrow \text{Attract}$
Objects even when they are separated.

For example:-

- ✓ When a comb is rubbed on dry hairs. It gets charged. It attracts bits of paper.
- ✓ When ebonite rod is rubbed with silk cloth, the ebonite rod is charged.

iii) **Magnetic Force**

- Two like magnetic poles repel, while two unlike magnetic poles attract each – other. The force between the magnetic poles is called the magnetic force.
Same poles (N – N / S – S) $\rightarrow$ Repel
Opposite poles (N – S / S – N) $\rightarrow$ Attract

For example:-

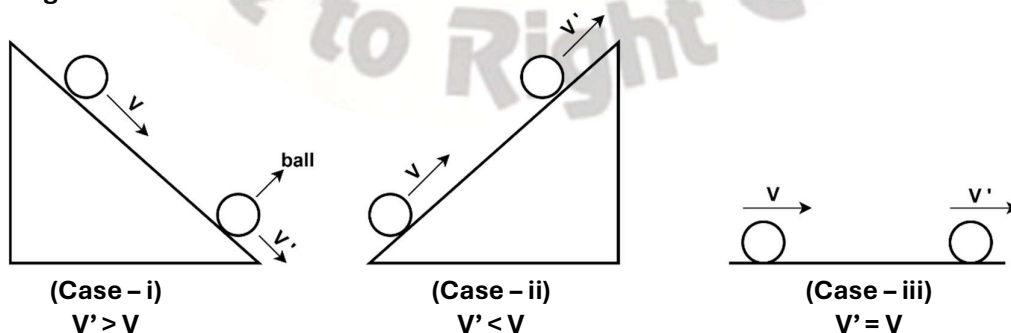
- ✓ Magnet attracts magnetic substances like nail, piece of iron etc.

• **GALILEO'S EXPERIMENTS**

- It was Galileo who recognised that a body moves with uniform velocity If no force acts on it.

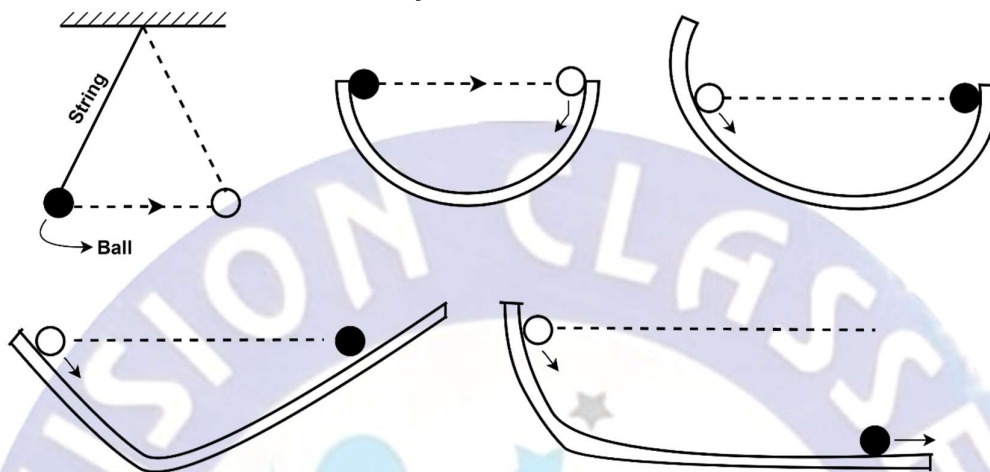
Before Galileo, there was a general belief that a force is necessary to keep a body move with uniform velocity.

Galileo arrived at the correct conclusion through a series of experiments and reasonings.



- **Conclusion**
- i) Moving up: Speed decreases
- ii) Moving down: Speed increases
- iii) Moving horizontally: Speed Constant

Galileo reached this conclusion in another way



Note:

1. ○ Released.
2. ● before stop.
3. If ball travels in horizontal surface, then it moves with uniform velocity.

- **Newton's laws of motion (N.L.M)**

- Newton proposed three laws of motion of an object (Either in rest or motion) which are known as Newton's law of motion.

1. Newton's first law of motion.
2. Newton's second law of motion.
3. Newton's third law of motion.

1. **Newton's first law of motion**

- Newton's first law of motion is defined as, "If a body at rest will remain at rest and if a body in motion will remain in uniform motion unless acted upon by an unbalanced force".

This law may also state as follows

- "A body remains unaccelerated if, and only if the resultant force on it is zero. i.e, the object is in equilibrium".

Special Note:

- Newton's first law of motion is also called Galileo's law of inertia because the idea was suggested by Galileo and was later formulated into a law by newton.

- Newton's first law is also called "law of inertia".

- **Inertia**

- The inability of a body to change on its state of rest or of uniform motion is known as inertia.

- **Inertia has three types**

- a) Inertia of rest.
- b) Inertia of motion.

c) Inertia of direction.

a) Inertia of rest

- If an object is in rest will remains at rest unless acted upon by an unbalanced force.

For example:-

- ✓ If the bus starts suddenly, we tend to fall backwards.
- ✓ The card – and – coin experiment.
- ✓ Striking a pile of Carrom coins.
- ✓ When a tree is vigorously shaken, some of the leaves fall from the tree.
- ✓ Falling down of dust particles while dusting a cloth.
- ✓ The trick of pulling a table cloth without disturbing the dishes on the table.

b) Inertia of motion

- If an object is in motion will remains at motion unless acted upon by an unbalanced force.

For example:-

- ✓ When the bus stops suddenly, then we – tends to fall upwards.
- ✓ A rolling ball.
- ✓ A person trying to get down from a running bus falls forward.
- ✓ Ripe fruits falls from the tree in the direction of wind.
- ✓ Satellite in space more difficult to stop moving.

c) Inertia of direction

- A body moving in a certain direction can't change its direction without any force being applied on it.

For example:-

- ✓ When a car makes a sharp turn at a high speed, the driver tends to get thrown to another side due to inertia of direction.
- ✓ The mud from the wheel of a moving vehicle files off tangentially.
- ✓ The spark coming out of a grinding stone is tangential to the rotating stone due to directional inertia.

Special Note:

- Inertia depends on mass thus, heavier the mass of body has greater inertia.

2. Newton's second law of motion

- The magnitude of the net force acting on a body is proportional to the product of the mass of the body and its acceleration. The direction of the force is the same as that of the acceleration.

i.e, $F \propto m$ (i)

& $F \propto a$ (ii)

On combining equation (i) and (ii)

$$F \propto ma$$

$$F = kma$$

Where, k is a constant

Or, Simply, $F = ma$

$$\Rightarrow m = F/a$$

i.e, $m \propto \frac{1}{a}$ [when F is constant] [Here, m = mass of body, a = acceleration created in body]

Conclusion:

- If same force (F) is applied on two different objects with different masses 10 kg and 50 kg.
So, the 10 kg mass object is more accelerates than 50 kg mass object.

Note:

1. $F_{\text{net}} = ma$, Thus the direction of acceleration and resultant force are same.
2. Internal forces cannot accelerates the object.
3. Mass is a property of object that opposes change in state of object. This property is called inertia of mass.
4. Greater the mass, greater is the inertia.
5. If total external force applied on a body is zero.
i.e, $\sum F_{\text{net}} = 0$
then, the object is in equilibrium condition/position/state
6. A single force cannot be balanced.
7. Two forces can be balanced if they have same magnitude and opposite direction.
8. Internal forces cannot change the state of object.
9. Internal forces are the forces by one part of the system on another part.

• **Linear momentum/momentum**

- The product of mass of a body and its instantaneous velocity is called momentum of that body.

If mass of the body = m

Instantaneous velocity = v

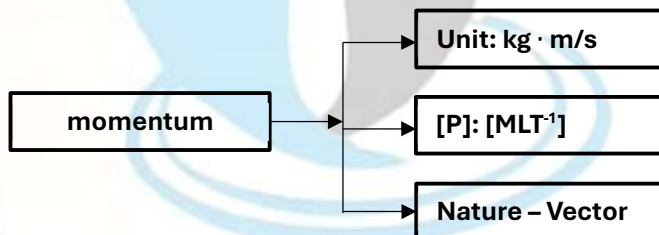
and linear momentum of body at that instant is P .

So, $P = m \cdot v$

Or momentum = mass $\times$ velocity

Note:

1. The momentum of a body is denoted by 'P'.



2. The direction of a linear momentum is same as the direction of velocity of object.
3. $P \propto m$
 $P \propto v$
i.e, heavier the mass or, greater the velocity the momentum is larger.

For example:-

- ✓ A duke's ball has more momentum than a Cosco ball if both have the same velocity.

Q. If a duke's ball of mass 155g thrown a velocity with 160 km/h. Find the momentum of the ball.

Ans $\because m = 155\text{g} = \frac{155}{1000} = 0.155\text{kg}$

$v = 160 \times \frac{5}{18} = \frac{800}{18} \text{ m/s}$

$\because P = m \cdot v$

$\because P = 0.155 \times \frac{800}{18}$

$\Rightarrow P = 6.89 \text{ kg m/s}$

▪ **Newton's Second law is defined in the term of momentum**

- The rate of change of momentum is equal to unbalance external force impressed on the object.

$$F_{net, \text{ external}} = \frac{\text{Change in momentum}}{\text{Change in time}}$$

- If the momentum at time t_1 is P_1 .
 & If the momentum at time t_2 is P_2 .

$$\therefore F_{net, \text{ external}} = \frac{P_2 - P_1}{t_2 - t_1}$$

$$\text{OR } F = \frac{mv - mu}{t}$$

Where,

v = final velocity of the object.

u = initial velocity of the object

So, The net force on a body is equal to the change its momentum per unit time.

Now check, $\frac{mv - mu}{t} \Rightarrow \frac{m(v - u)}{t}$

$$\Rightarrow m \times \frac{v - u}{t} = m \times a$$

$$= F$$

$$\left(\begin{array}{l} \because v = u + at \\ \Rightarrow a = \frac{v - u}{t} \end{array} \right)$$

- **1 Newton is defined as**

- If a force acting on a body of mass 1kg produces an acceleration of 1m/s^2 in it, then the force is called 1 Newton.

$$\therefore F = ma$$

$$\therefore 1\text{N} = 1\text{Kg} \times 1\text{m/s}^2$$

$$\Rightarrow 1\text{N} = 1\text{Kg m/s}^2$$

The S.I unit of force is 'Newton' which is denoted by 'N'.

- **System of particles/objects/bodies**

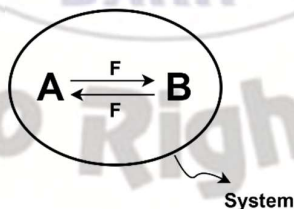
- A system is collection/group of two or more objects

For example:-

- ✓ Train, Tonga, bullock – cart etc.

- **Internal forces of a System**

- Let A and B two objects in a system, the force exerted by A on B and the force exerted by B on A are called internal forces.



- **External Forces**

- Any Force applied on either A or B by an external particle (other than A & B) is called an external force.

- **The momentum of a System**

- If n^{th} objects/bodies in a system moving along same straight line then the momentum of the system is arithmetic sum of their momenta (momentum of each particle).

$$\text{i.e, } P = m_1v_1 + m_2v_2 + \dots + m_nv_n.$$

where,

$m_1, m_2, \dots, m_n$ are masses of objects.

&

$v_1, v_2, \dots, v_n$ are velocities of objects.

Note:

1. Internal force can change only the momenta of different objects in the system but not the total momentum of the system.
2. According to Newton's first law, an object remains at rest or moves with a constant velocity, if the total force acting on it zero. In this case momentum is constant and this law is called the principle of conservation of linear momentum.

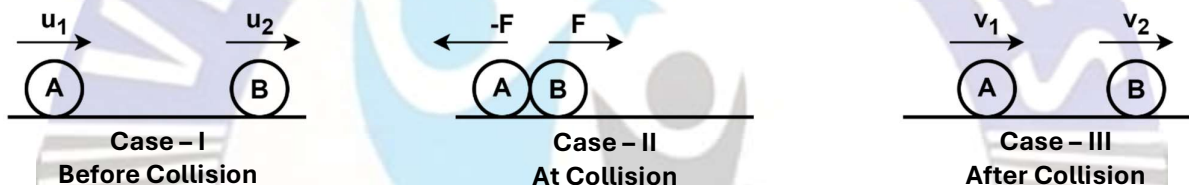
• **Verification of principle of conservation of linear momentum**

- If the net external force on a system of particles is zero, the linear momentum of the system remains constant.

Proof:

- Let two objects A and B of masses m_1 and m_2 kg are kept on a smooth surface that friction can be neglected. If velocity of A and B are u_1 & u_2 .

If velocities: $u_1 > u_2$



Case 1: Before the collision

The linear momentum of object A = $m_1 u_1$

The linear momentum of object B = $m_2 u_2$

Let A and B in a system. If the linear momentum of the system is P

So, $P = m_1 u_1 + m_2 u_2 \dots\dots\dots(i)$

Case 2: Let A and B collides for a short time t_c

Time of collision

According to Newton's third law,

- A will push B towards the right, and B will push A towards the left with equal magnitude of Force F.
 ➤ Take right direction is +ve and left direction is -ve.
 ➤ So, the force applied on B by A is F and applied on A by B is -F.

Case 3: After Collision of A & B

- Due to exert a force both objects are separated to each other and gets final velocity v_1 and v_2 .
 Now, object A exert force (-F) by B produces an acceleration (a_1) in A (object).

$$\therefore a_1 = \frac{-F}{m_1}$$

$$\text{Or, } \frac{v_1 - u_1}{t_c} = \frac{-F}{m_1}$$

$$\Rightarrow m_1(v_1 - u_1) = -F \cdot t_c$$

$$\Rightarrow m_1 v_1 - m_1 u_1 = -F \cdot t_c \dots\dots\dots(ii)$$

Similarly, Force on B is F and B accelerates by a_2 .

$$\Rightarrow a_2 = \frac{F}{m_2}$$

$$\Rightarrow \frac{v_2 - u_2}{t_c} = \frac{F}{m_2}$$

$$\Rightarrow m_2(v_2 - u_2) = F \times t_c$$

$$\Rightarrow m_2 v_2 - m_2 u_2 = F \times t_c \dots\dots\dots(iii)$$

$\therefore$ F and -F are equal in magnitude

$\therefore$ adding equation (ii) & (iii)

$$\Rightarrow m_1 v_1 - m_1 u_1 + m_2 v_2 - m_2 u_2 = -F \times t_c + F \times t_c$$

$$\Rightarrow m_1 v_1 + m_2 v_2 = m_1 u_1 + m_2 u_2$$

$$\Rightarrow m_1 v_1 + m_2 v_2 = P \quad [\text{from equation (i)}]$$

$$\Rightarrow P = m_1 v_1 + m_2 v_2$$

Thus, linear momentum of the system remains constant.

▪ **Application of Newton's Second Law**

- i) Hand moves back during on catching a fast cricket ball.
- ii) In the case of a high jump or, long jump Athlete is provided a heap of sand on the ground to fall upon.
- iii) Use of seat belt in cars.
- iv) Pushing a car and a truck.
- v) Kick a ball.

Note:

1. A rocket works on the principal of conservation of momentum.
2. A jet aeroplane also works on the perpendicular of conservation of momentum.

• **Impulse**

- The change in momentum is called impulse.
 i.e, Impulse = Final momentum – initial momentum

The momentum of body is denoted by 'y'. It is a vector quantity.

$$\text{Thus, } \vec{j} = \vec{P}_{final} - \vec{P}_{initial}$$

$$\Rightarrow \vec{j} = m\vec{v} - m\vec{u} = m(\vec{v} - \vec{u})$$

$$\Rightarrow \vec{j} = \frac{m(\vec{v} - \vec{u})}{\Delta t} \times \Delta t$$

$$\Rightarrow \vec{j} = m \times \frac{\vec{v} - \vec{u}}{\Delta t} \times \Delta t$$

$$\Rightarrow \vec{j} = m \times \vec{a} \times \Delta t \quad [\because \vec{a} = \frac{\vec{v} - \vec{u}}{t}]$$

$$\Rightarrow \vec{j} = F \times \Delta t \quad [\Delta t = t_2 - t_1]$$

As a result:

- For a constant force the product of force applied, and time taken is called impulse.

3. Newton's third law of motion

- Newton's third law of motion is defined as, "To every action there is equal and opposite reaction".
 In other words,

In any interaction between two bodies the force applied by the first body on the second is equal and opposite to the force applied by the second body on the first.

▪ **Application of Newton's third law**

- | | |
|------------------------------------|--|
| i) In Firing a bullet from a gun. | vii) In the case of bullock – cart, Horse – cart (Tamtum) etc. |
| ii) When we walking on the ground. | viii) In the case of deflating balloon. |
| iii) In the case of swimming. | ix) In the case of rocket launch. |
| iv) In the case of boating. | x) In the case of flying a jet plane. |
| v) In the case of ship. | |
| vi) When a man step out of a boat. | |